Programming Robots 

Brownie Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

What is a robot?

A robot is a machine that can: - **SENSE** — see, hear, or feel things around it 👀 - **THINK** — decide what to do 🧠 - **ACT** — move or do a job 🦾

Question for you: What robots have you seen in real life?

What is a PROGRAM?

A program is a **list of steps** that tells a robot exactly what to do.

Like a recipe!  1. Get a bowl 2. Add flour 3. Add eggs 4. Mix!

Robots follow their recipe **exactly** — even if it's wrong!

Game Time: Robot Says! 🎮

Your mentor is now a ROBOT.

- Robots only follow **clear** commands
- “Take 3 steps forward” ✅
- “Go over there” ❌ *ERROR BEEP!*

Your mission: program your mentor to walk across the room!

The Human Maze 🗺️

Work in pairs: - One of you is the **ROBOT** 🤖 - One of you is the **PROGRAMMER** 👩‍💻

1. Programmer: line up your arrow cards
2. Robot: follow the cards EXACTLY
3. Hit an obstacle? That's a **BUG**! 🐛
4. Fix your cards and try again

Bugs are GOOD! 

When a program doesn’t work, that’s called a **bug**.
Real programmers find bugs **every single day**.
Finding a bug means you’re doing real programming!
Plan → Run → Find the bug → Fix it → Run again

Now let’s program on a computer! 💻

We’ll use **MakeCode Arcade**

You will: 1. Make a character (a “sprite”) 2. Make it move with arrow keys 3. Collect a star ⭐ 4. Play a sound 🔊

Follow along with your mentor, one block at a time!

Show and Tell ✨

Show a friend ONE thing your program does!

Then answer: - What did you FIX when it didn’t work? - What would you add next?

You did it! 🎉

Today you: - ✅ Learned what a program is - ✅ Programmed a human robot - ✅ Fixed bugs like a real programmer - ✅ Made a computer program in MakeCode Arcade

You are a programmer now!

Keep going at home! 🏠

- **MakeCode Arcade:** arcade.makecode.com (free!)
- **Scratch Jr.** app — make stories and games
- **Code.org** — Hour of Code activities
- Ask your troop leader about the next robotics badge!